Yihong Liu

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Education

University of California, Santa Barbara — Expected June 2027
Ph.D. in Economics — *Santa Barbara, CA*

Duke University — June 2021
M.A., Economics — *Durham, NC*

Shanghai University of Finance and Economics — June 2019
B.A., Economics — *Shanghai, China*

Research Interest

Research Interest:
Econometrics, Applied Econometrics, Environmental and Natural Resource Economics

Working Papers

Matched Triple Differences: A Framework for Valid Covariate Adjustment (Job Market Paper)
Abstract: A common empirical strategy in triple-differences (DDD) is to include either covariate trends or covariate levels in a three-way fixed effects (3WFE) regression. This strategy is typically motivated by the conditional parallel gaps assumption which assumes that deviations from parallel trends are similar among units with comparable observed covariates. We formally study both 3WFE specifications and show that, in general, neither consistently estimates the average treatment effect on the treated (ATT). Our diagnosis has two parts. First, we show that the OLS estimands of both specifications fail to satisfy the covariate balancing condition that is sufficient for them to equal the ATT. Second, we characterize the additional assumptions for each OLS estimand to equal the ATT. To address these limitations, we propose a matched triple-differences framework that accommodates a general class of matching procedures, including nearest-neighbor matching and kernel matching. Within this framework, we construct a class of consistent matching estimators, establish their asymptotic normality, and provide a consistent variance estimator that accounts for the variability introduced by the matching step. Our empirical application shows that the proposed matched triple-differences estimator can yield estimates and qualitative conclusions that differ from those obtained using the 3WFE regression.

Post Matching Two-way Fixed Effects Estimation (with Gonzalo Vazquez-Bare)
Abstract: Identification in difference-in-differences (DiD) settings is often justified under a conditional parallel trends assumption, under which units with similar observed covariates are expected to have followed similar outcome trends in the absence of treatment. In such settings, researchers often combine matching with two-way fixed effects (2WFE) methods. Specifically, in a first step, each treated unit is matched to one or more never-treated units based on observed pre-treatment covariates. In the second step, treatment effects are estimated with a 2WFE regression in the matched sample, reweighting the never-treated units by the number of times they are matched. We formally analyze this common practice and highlight three problems. First, with variation in treatment timing, the post-matching 2WFE estimator that pools all treated cohorts has an asymptotic "pooling" bias, even when the treatment effect is constant across units and over time.

Second, failing to account for the variability introduced by the matching procedure yields invalid standard error estimators, which can be biased upwards or downwards depending on the data generating process. Third, imperfect matches generate a matching bias that can be first-order for inference, but the required bias correction to address it is commonly omitted when using post-matching 2WFE estimators in empirical work. Thus, conventional t-statistics and confidence intervals based on the uncorrected estimator may be invalid. We provide consistent, asymptotically normal bias-corrected matched DiD estimators of treatment effects, event-study coefficients and other aggregate magnitudes, with consistent standard errors that account for the matching step. We illustrate our results with simulations and with two empirical applications.

Estimating Resource Games with Nonlinear Dynamics (with Shuchen Tsao, Submitted)
Abstract: Many policy contexts, from fisheries management to epidemic control, are governed by nonlinear reproduction dynamics and by strategic interaction among multiple decision makers. This combination has historically made structural estimation computationally forbidding. We develop a tractable Generalized Method of Moments (GMM) framework for dynamic resource games with heterogeneous agents and fully nonlinear, interactive state dynamics. We provide sufficient conditions under which the Markov-perfect Nash equilibrium admits analytical optimality conditions, and establish consistency and asymptotic normality of the resulting GMM estimator. Applying the framework to the Japanese anchovy fishery shared by China, Japan, and Korea, we recover each country's implicit ecological valuation and simulate the equilibrium effect of climate-driven shifts in fish migration patterns. We further show that linearizing the equations of motion, the standard route to tractability, substantially understates these ecological values.

Work in Progress

Identification and Inference on Treatment Effects under Covariate-Adaptive Randomization and Sample Selection

Conference and Seminar

2026: University of California, Santa Barbara (CARE), AERE at the Western Economic Association International (WEAI)

2025: University of California, Santa Barbara (AMEL)

2024: CES North America Conference

Teaching Experience

University of California, Santa Barbara

Econ 241A Econometrics I (Ph.D. Core)	Fall 2024, 2025
Econ 241B Econometrics II (Ph.D. Core)	Winter 2023, 2024, 2026
Econ 210A Microeconomics I (Ph.D. Core)	Fall 2022, 2023
Econ 140C Machine Learning and AI	Fall 2026
Econ 140A Introduction to Econometrics I	Spring 2023, 2025
Econ 2 Introductory Macroeconomics	Spring 2022
Econ 1 Introductory Microeconomics	Fall 2021, Winter 2022

Awards & Honors

Job Market Fellowship 2026
University of California, Santa Barbara

Research Quarter Fellowship 2025, 2026
University of California, Santa Barbara

Outstanding Graduate Course Teaching Assistant of the Year 2024
University of California, Santa Barbara

Gretler Fellowship 2024
University of California, Santa Barbara

M.A Merit Scholar Award
Duke University *2019, 2020*

Renmin Scholarship of the Graduating Class: Third Prize
Shanghai University of Finance and Economics *2019*

Outstanding Exchange Student of SUFE: Second Prize
Shanghai University of Finance and Economics *2018*

Specialized Skills

Programming Languages: STATA, R, MATLAB
Languages: Chinese(native), English(fluent)

References

Douglas Steigerwald
Professor of Economics
University of California, Santa Barbara
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Andrew Plantinga
Professor of Natural Resource Economics
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Max Farrell
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Gonzalo Vazquez-Bare
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